

Q1 Design a class Hierarchy (pseudocode)

CLASS Person

DATA name, age

METHOD readPerson()

Read name, age

END METHOD

END CLASS

CLASS STUDENT Inherits Person

DATA regNo, marks[5], average

METHOD CalculateAverage()

SUM $\leftarrow$ 0

FOR $i \leftarrow 0$ TO 4

SUM $\leftarrow$ SUM + marks[i]

END FOR

average $\leftarrow$ SUM / 5

END METHOD

END CLASS

METHOD displayDetails()

Print name

Print age

Print regNo

Print average

Print researchArea

END METHOD

END CLASS

2) Complete the missing Pseudocode

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$

$A[j] \leftarrow -1$

END IF

END FOR

END FOR

3) Debug the Pseudocode

RETURN length + length

4) Trace the Execution

Sum = 0

$i = 1 \rightarrow \text{Sum} = -1$

$i = 2 \rightarrow \text{Sum} = 1$

$i = 3 \rightarrow \text{Sum} = -2$

$i = 4 \rightarrow \text{Sum} = 2$

$i = 5 \rightarrow \text{Sum} = -3$

output = -3

5) Write Efficient pseudocode

```
BEGIN  
READ N  
READ ARRAY A  
largest  $\leftarrow -\infty$   
Second  $\leftarrow -\infty$   
  
FOR  $i \leftarrow 0$  TO  $N-1$   
  IF  $A[i] > \text{largest}$  THEN  
    Second  $\leftarrow$  largest  
    largest  $\leftarrow A[i]$   
  ELSE IF  $A[i] > \text{Second}$  AND  $A[i] \neq \text{largest}$  THEN  
    Second  $\leftarrow A[i]$   
  END IF  
END FOR  
PRINT SECOND  
END
```

6) Convert Problem Statement into Pseudo code

```
INTERFACE Payment  
METHOD Payment  
END  
  
CLASS CreditCard IMPLEMENTS Payment  
CLASS UPI IMPLEMENTS Payment  
CLASS NETBanking IMPLEMENTS Payment  
  
BEGIN  
  READ choice  
  CALL Pay (amount)  
END
```


7) Fill in the blanks

Inherits

New Dog()

8) Find Logical Error

```
CONSTRUCTOR (id, name)
```

```
  this.id ← id
```

```
  this.name ← name
```

```
END CONSTRUCTOR
```

9) Case Study

```
CLASS Vehicle
```

```
  DATA VehicleNo, owner name
```

```
  Method display ()
```

```
END
```

```
CLASS Car INHERITS Vehicle
```

```
  DATA FuelType
```

```
END
```

```
CLASS Truck Inherits Vehicle
```

```
  DATA load capacity
```

```
END
```

```
BEGIN
```

```
  CREATE objects
```

```
  STORE objects
```

```
  Display objects
```

```
END
```